

Sequential Tilt-Motion Lock



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Here is a project for locking and unlocking with a tilt sensor by tilting it in a defined sequence. It uses an accelerometer module to detect the tilt motion. If the sequence matches with the predefined motion sequence, the lock opens. You can build this lock for a briefcase, ballot box, portable cashbox or even as a door-lock using suitable mechanical arrangement.

Circuit and working

Fig. 1 shows circuit diagram of the sequential tilt-motion lock. The circuit is built around Arduino Uno

board (Board1), accelerometer module (ACC.1), solenoid lock/electric strike and a few other components.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino Uno is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable, or power it

with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

Pins A0, A1, A2, A3, A4 and A5 of Board1 are connected to pins ST, Z-axis, Y-axis, X-axis, GND and Vcc of the accelerometer module, respectively. Pin 10 of Board1 is connected to solenoid driver transistor through which the solenoid lock is connected.

Accelerometer module. An accelerometer is an electromechanical device that measures acceleration. The accelerometer module used here is based on ADXL335 triple-axis accelerometer from Analog Devices. The sensor has a full sensing range of $\pm 3g$.

The microcontroller in Board1 receives data at pins A1, A2 and A3 for z, y and x axes, respectively, from the accelerometer. This data is continuously compared by Board1 with predefined values for each axis. If the received sequence matches, Board1 unlocks the lock, which is either a solenoid lock or any other suitable magnetic lock. (We used a 12V electric strike for testing.) Glowing of LED1 indicates that the lock is open.

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Uno board
- ACC.1 - ADXL335 3-axis accelerometer module
- LED1 - 5mm LED
- T1 - BC548 npn transistor
- D1 - 1N4007 rectifier diode

Resistors (all $\frac{1}{4}$ -watt, 5% carbon):

- R1, R2 - 1-kilo-ohm

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- S1 - On/off switch
- BATT.1 - 12V battery
- 12V electric strike (solenoid lock)

Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	High
TP3	Low

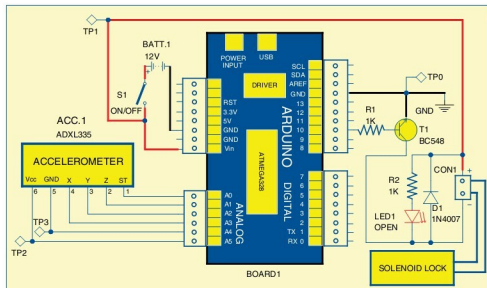


Fig. 1: Circuit diagram of sequential tilt-motion lock

Software

The software for this project is written in Arduino programming language. The Arduino Uno is programmed using Arduino IDE software. ATmega328 on Arduino Uno comes with a boot loader that allows you to upload new code to it without the use of external hardware programmer. It communicates using the STK500 protocol. You can also bypass the boot loader and program the microcontroller through in-circuit serial programming (ICSP) header, but boot loader programming is quicker and easier. Select the correct board from 'Tools → Board' in Arduino IDE and burn